

Scenario Analyses: Effect of Utility Assumptions

METIMMOX-1 Economic Evaluation

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1 Overview

This scenario analysis examines the impact of utility assumptions on the cost-effectiveness of biomarker-guided immunotherapy strategies for metastatic MSS/pMMR colorectal cancer.

1.1 Utility Scenarios

We compare three utility scenarios:

1. **Base Case:** Standard utility values ($u_{np} = 0.9077$, $u_p = 0.9005$)
2. **Scenario 1 - Unadjusted Life Years:** All utilities set to 1.0 (no quality adjustment)
3. **Scenario 2 - Alternative Utilities:** Lower utility values ($u_{np} = 0.73$, $u_p = 0.59$)

1.2 Model Structures

Each scenario is analyzed using three alternative model structures for the survival analysis:

Table 1: Model Structures for Survival Analysis

Model	Name	Formula Pattern	Description
Model A	Joint (All-in-One)	Age + sex + Rx + crp*Rx + tlr*Rx + tmb_braf*Rx	All biomarkers and all treatment interactions in a single model
Model B	Focused	Age + sex + Rx + crp + tmb_braf + crp:Rx + tmb_braf:Rx	Per-strategy formulas: CRP/TMB_BRAF with both interactions, TLR with TLR-only interaction
Model C	Separate	Age + sex + Rx + [biomarker]:Rx	Only one biomarker and its treatment interaction per model

For each scenario, we present deterministic cost-effectiveness results comparing all three model structures, followed by probabilistic sensitivity analysis.

2 Base Case: Standard Utility Values

This analysis uses standard utility values derived from the IPD dataset ($u_{np} = 0.9077$, $u_p = 0.9005$).

Table 2: Base Case Utility Parameter Values

Parameter	Value	Description
u_{np} (Non-progressed utility)	0.9077	Utility value in non-progressed disease state
u_p (Progressed utility)	0.9005	Utility value in progressed disease state

2.1 Multi-Model Comparison

Table 3: Base Case Cost-Effectiveness Results by Model Structure

Strategy	Model A: Joint			Model B: Focused			Model C: Separate
	Cost (A)	QALY (A)	ICER (A)	Cost (B)	QALY (B)	ICER (B)	
Standard of Care	\$20,891	1.846	Reference	\$21,261	1.940	Reference	\$21,261
CRP-guided	\$54,894	1.783	\$1,949,087	\$55,118	1.897	\$14,980,066	\$55,553
TLR-guided	\$84,959	1.702	Reference	\$86,411	1.699	Reference	\$88,251
TMB/BRAF-guided	\$63,561	1.868	Reference	\$61,542	1.942	Reference	\$67,926

Table 4: Optimal Strategy by Model Structure - Base Case

Model Structure	Optimal Strategy	NMB at WTP=EUR 51,000
Model A: Joint	Standard of Care	\$73,254
Model B: Focused	Standard of Care	\$77,657
Model C: Separate	Standard of Care	\$77,657

2.2 Cost-Effectiveness Planes

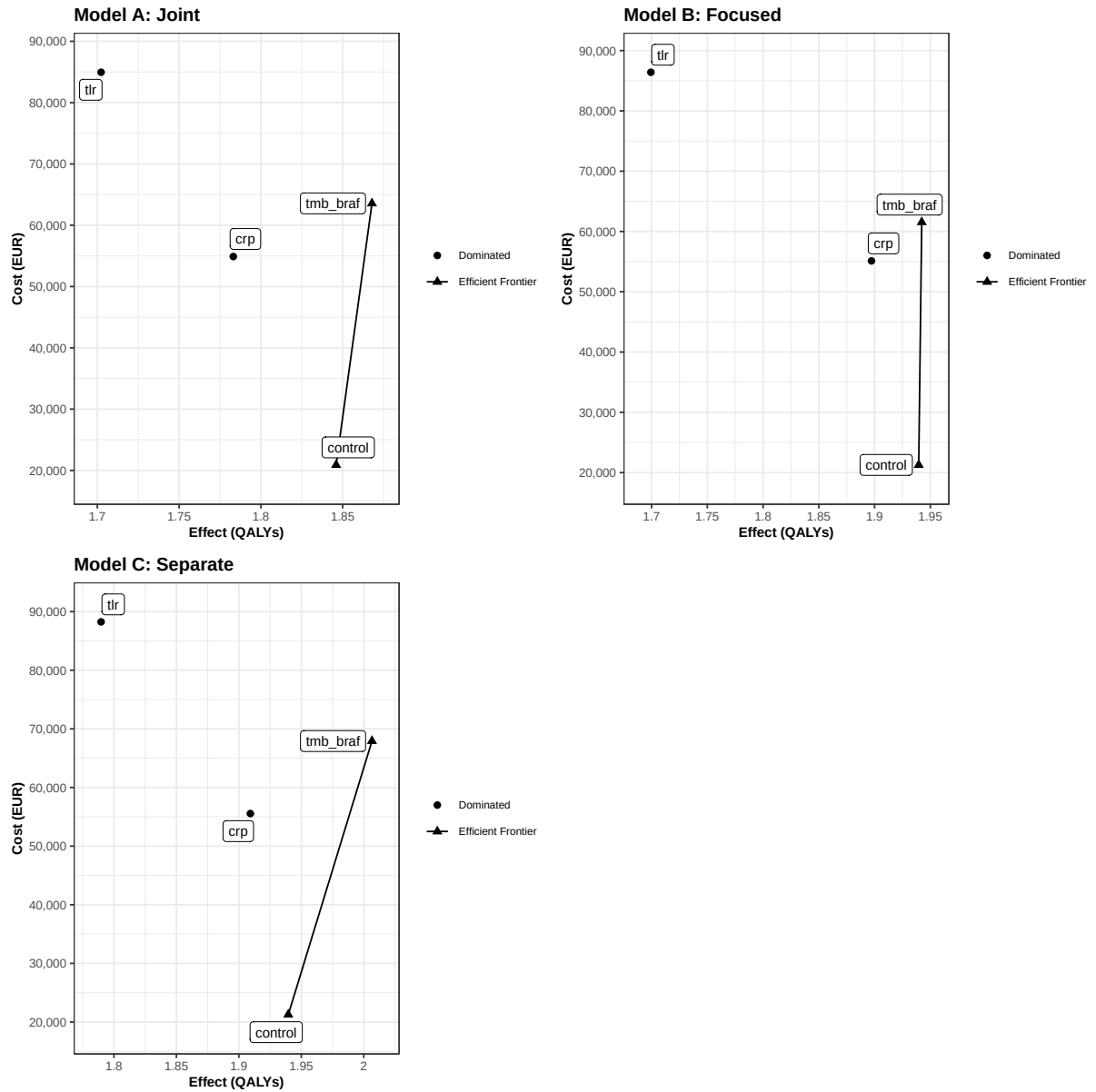


Figure 1: Cost-Effectiveness Planes by Model Structure - Base Case

2.3 Probabilistic Sensitivity Analysis (Model A)

PSA results are presented for Model A (Joint) only, as this is the primary model structure. The deterministic results above show comparison across all three model structures.

Table: PSA Summary Statistics - Base Case (Model A)

Table 5: PSA Summary - Base Case (Model A: Joint)

Strategy	meanCost	meanEffect	sdCost	sdEffect
control	21172.11	1.78	2893.92	0.36
crp	54804.86	1.81	8973.41	0.27
tlr	85012.92	1.76	16864.06	0.31

tmb_braf	63211.07	1.93	10767.13	0.36
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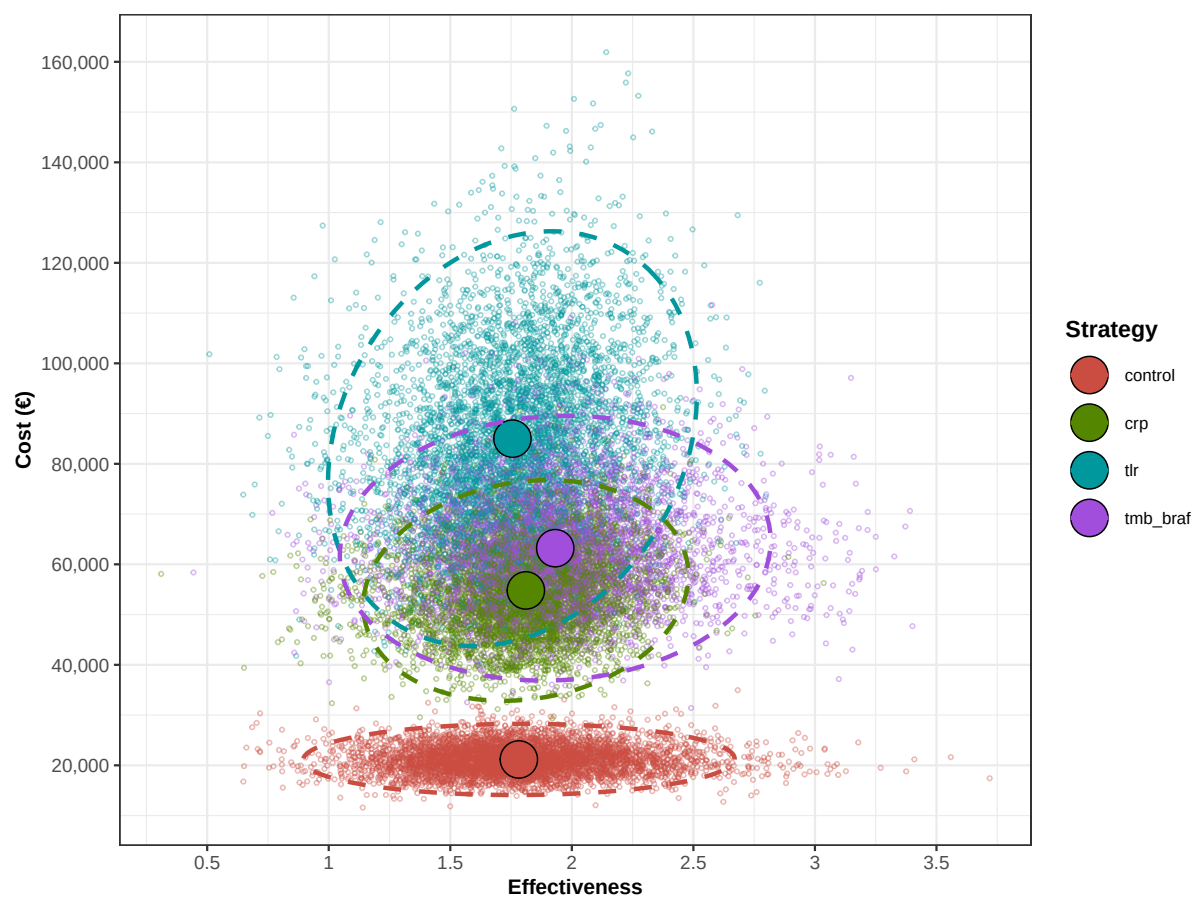


Figure 2: PSA Scatterplot - Base Case (Model A: Joint)

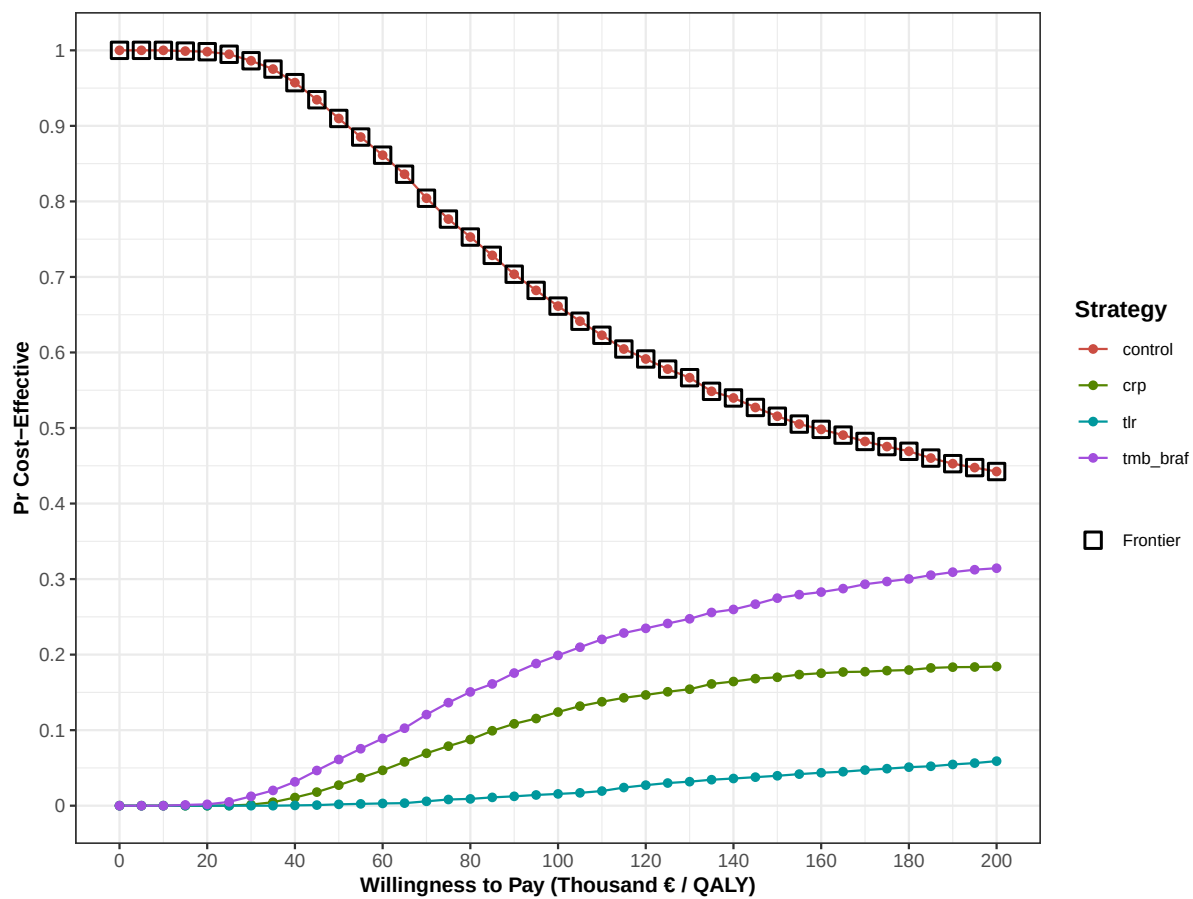


Figure 3: Cost-Effectiveness Acceptability Curve - Base Case (Model A: Joint)

3 Scenario 1: Unadjusted Life Years (All Utilities = 1.0)

In this scenario, we analyze life years gained without quality-of-life adjustments, providing insight into the survival benefit independent of quality-of-life considerations.

Table 6: Utility Parameter Modifications for Scenario 1

Parameter	Base Case	Scenario 1	Difference
u_np (Non-progressed utility)	0.9077	1	0.0923
u_p (Progressed utility)	0.9005	1	0.0995

3.1 Multi-Model Comparison

Table 7: Scenario 1 Cost-Effectiveness Results by Model Structure (Unadjusted Life Years)

Strategy	Model A: Joint			Model B: Focused			Model C: Separate	
	Cost (A)	LY (A)	ICER (A)	Cost (B)	LY (B)	ICER (B)	Cost (C)	LY (C)
Standard of Care	\$20,891	2.043	Reference	\$21,261	2.144	Reference	\$21,261	2.144
CRP-guided	\$54,894	1.971	\$1,894,792	\$55,118	2.097	\$14,290,720	\$55,553	2.111
TLR-guided	\$84,959	1.881	Reference	\$86,411	1.877	Reference	\$88,251	1.971
TMB/BRAF-guided	\$63,561	2.065	Reference	\$61,542	2.147	Reference	\$67,926	2.211

Table 8: Optimal Strategy by Model Structure - Scenario 1

Model Structure	Optimal Strategy	NMB at WTP=EUR 51,000
Model A: Joint	Standard of Care	\$83,298
Model B: Focused	Standard of Care	\$88,108
Model C: Separate	Standard of Care	\$88,108

3.2 Cost-Effectiveness Planes

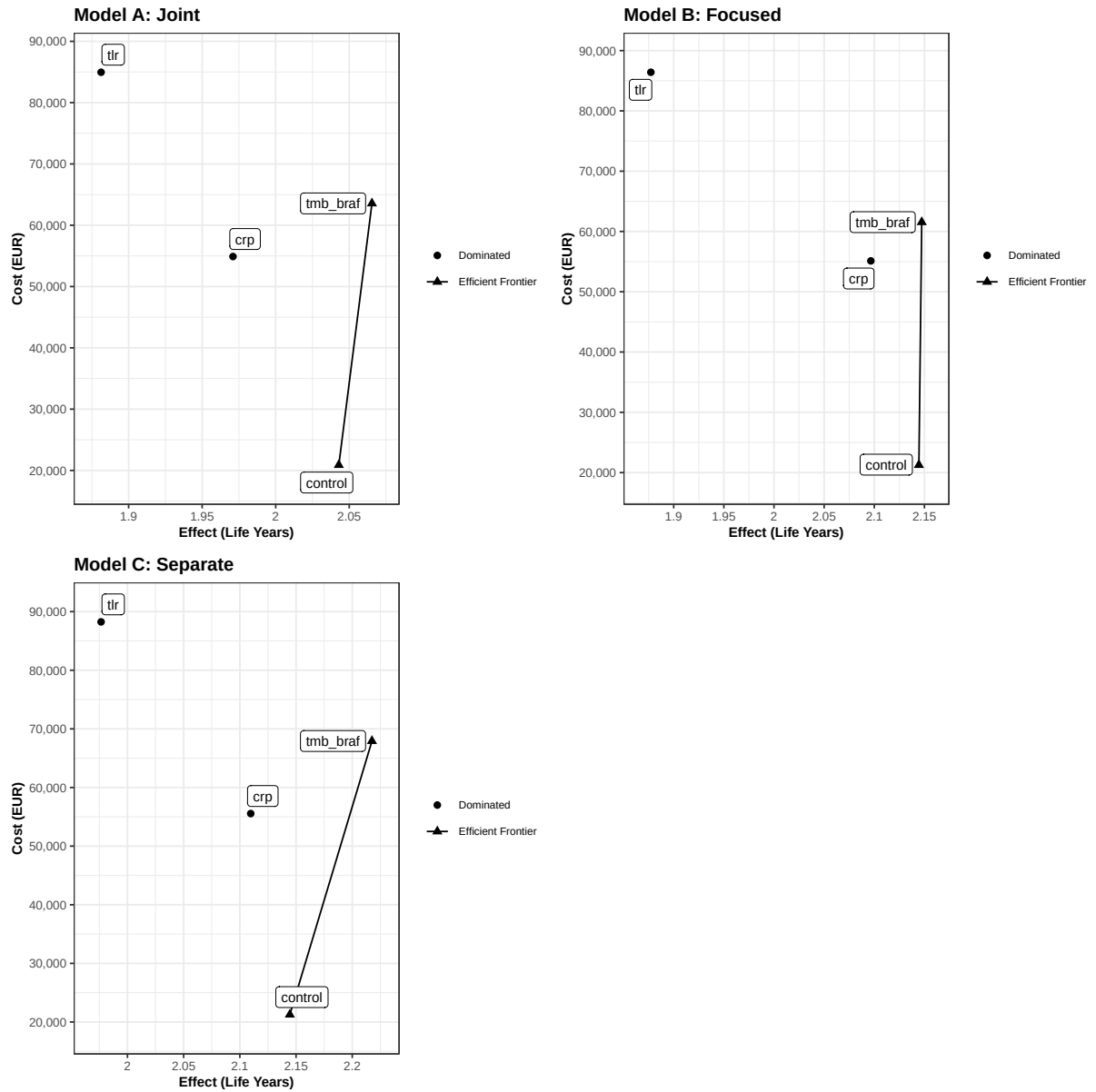


Figure 4: Cost-Effectiveness Planes by Model Structure - Scenario 1 (Unadjusted Life Years)

3.3 Probabilistic Sensitivity Analysis (Model A)

PSA results are presented for Model A (Joint) only.

Table: PSA Summary Statistics - Scenario 1 (Model A)

Table 9: PSA Summary - Scenario 1 (Model A: Joint, Unadjusted Life Years)

Strategy	meanCost	meanEffect	sdCost	sdEffect
control	21208.57	1.97	2937.94	0.33
crp	55038.10	2.00	9166.24	0.20
tlr	85329.52	1.94	16657.83	0.26
tmb_braf	63470.39	2.14	11131.45	0.33

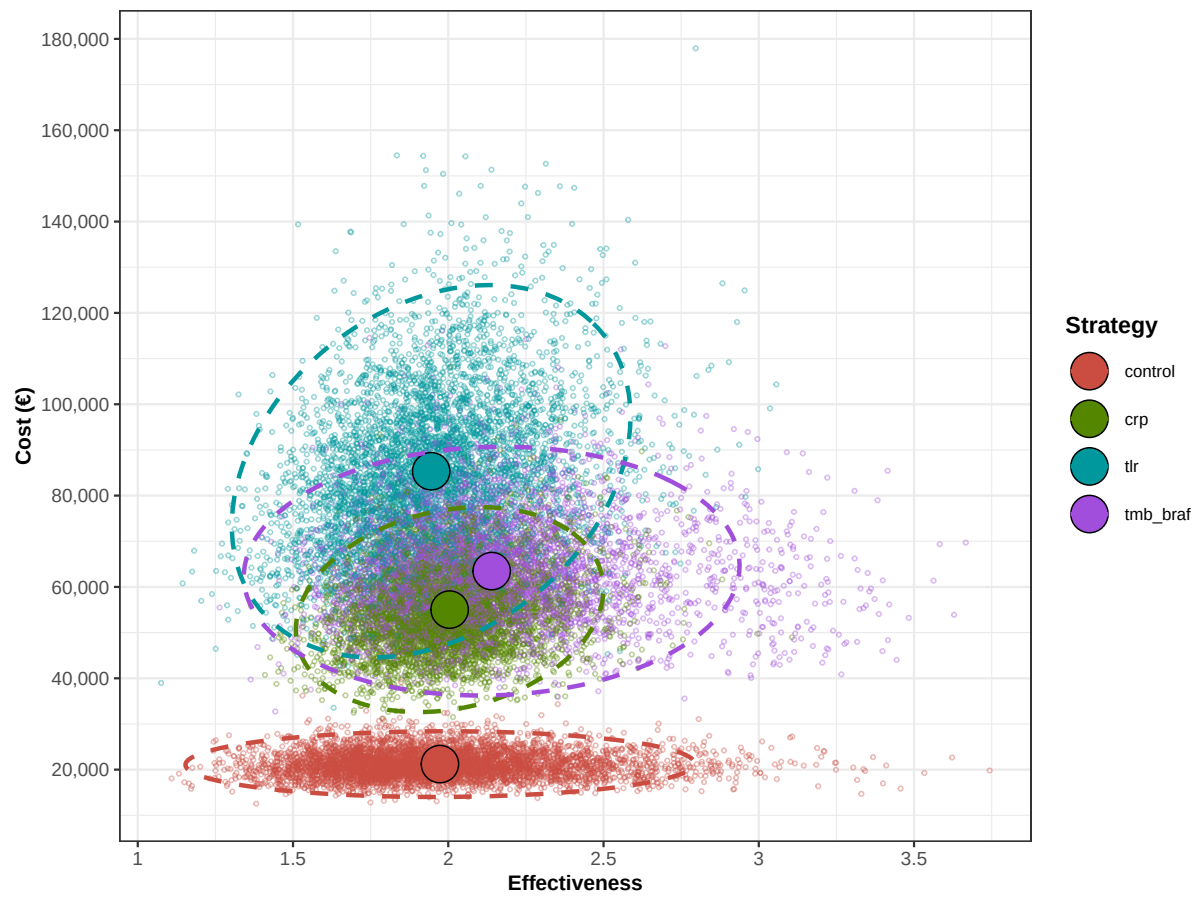


Figure 5: PSA Scatterplot - Scenario 1 (Model A: Joint, Unadjusted Life Years)

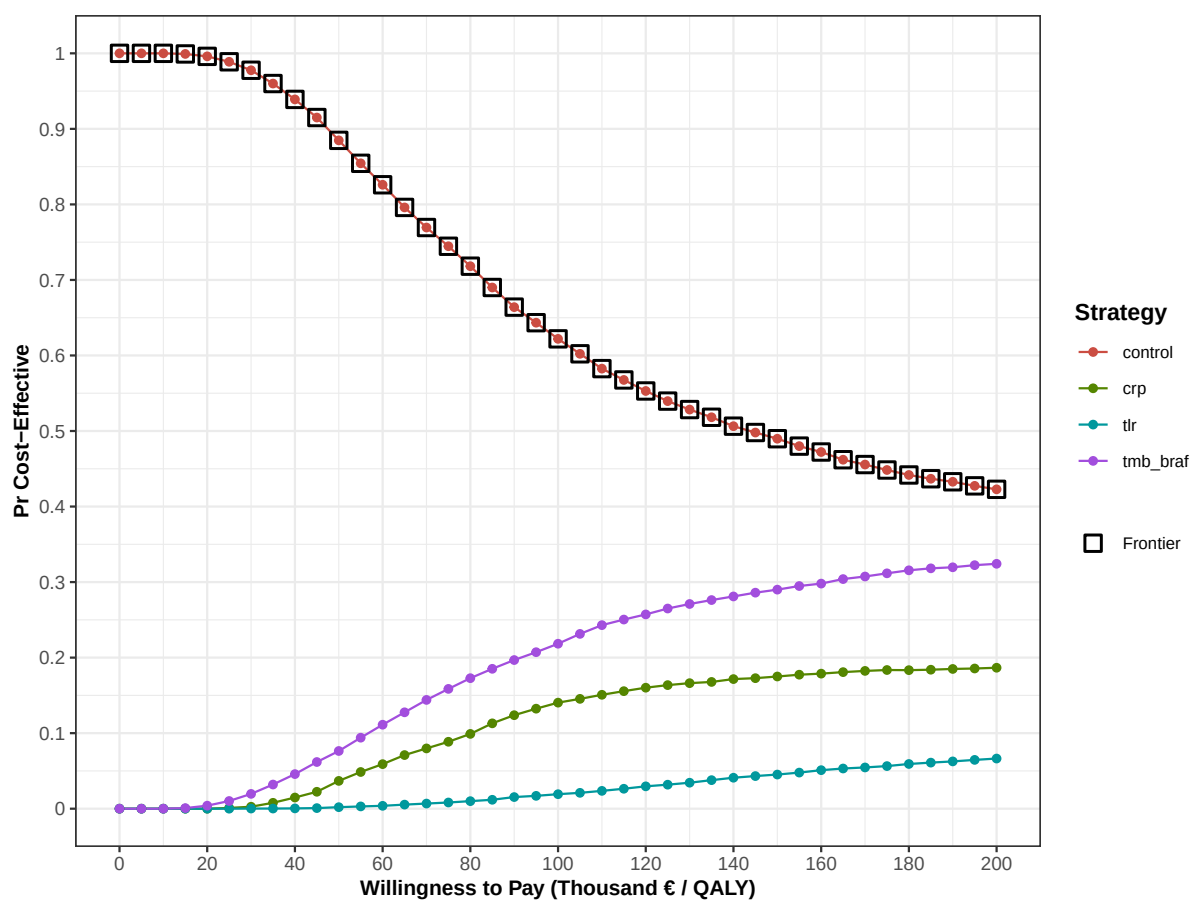


Figure 6: Cost-Effectiveness Acceptability Curve - Scenario 1 (Model A: Joint)

4 Scenario 2: Alternative Utility Values ($u_{np} = 0.73$, $u_p = 0.59$)

This scenario examines the cost-effectiveness under alternative, lower utility assumptions that may reflect more conservative estimates of quality of life. They are derived from the EQ-5D index score of the CORRECT trial (Grothey et al. Lancet 2013, as cited by Gourzoulidis et al. J Comp Effect Res 2018)-

Table 10: Utility Parameter Modifications for Scenario 2

Parameter	Base Case	Scenario 2	Difference	% Change
u_{np} (Non-progressed utility)	0.9077	0.73	-0.1777	-19.5770
u_p (Progressed utility)	0.9005	0.59	-0.3105	-34.4808

Beta Distribution Parameters for PSA:

Table 11: Beta Distribution Parameters for Scenario 2 PSA

Parameter	Mean	CV	Alpha	Beta
u_{np}	0.73	0.15	11.270	4.168
u_p	0.59	0.15	17.632	12.253

4.1 Multi-Model Comparison

Table 12: Scenario 2 Cost-Effectiveness Results by Model Structure (Alternative Utilities)

Strategy	Model A: Joint			Model B: Focused			Model C: Separate
	Cost (A)	QALY (A)	ICER (A)	Cost (B)	QALY (B)	ICER (B)	Cost (C)
Standard of Care	\$20,891	1.328	Reference	\$21,261	1.430	Reference	\$21,261
CRP-guided	\$54,894	1.324	\$955,489	\$55,118	1.417	\$8,767,657	\$55,553
TLR-guided	\$84,959	1.270	Reference	\$86,411	1.281	Reference	\$88,251
TMB/BRAF-guided	\$63,561	1.373	Reference	\$61,542	1.434	Reference	\$67,926

Table 13: Optimal Strategy by Model Structure - Scenario 2

Model Structure	Optimal Strategy	NMB at WTP=EUR 51,000
Model A: Joint	Standard of Care	\$46,848
Model B: Focused	Standard of Care	\$51,663
Model C: Separate	Standard of Care	\$51,663

4.2 Cost-Effectiveness Planes

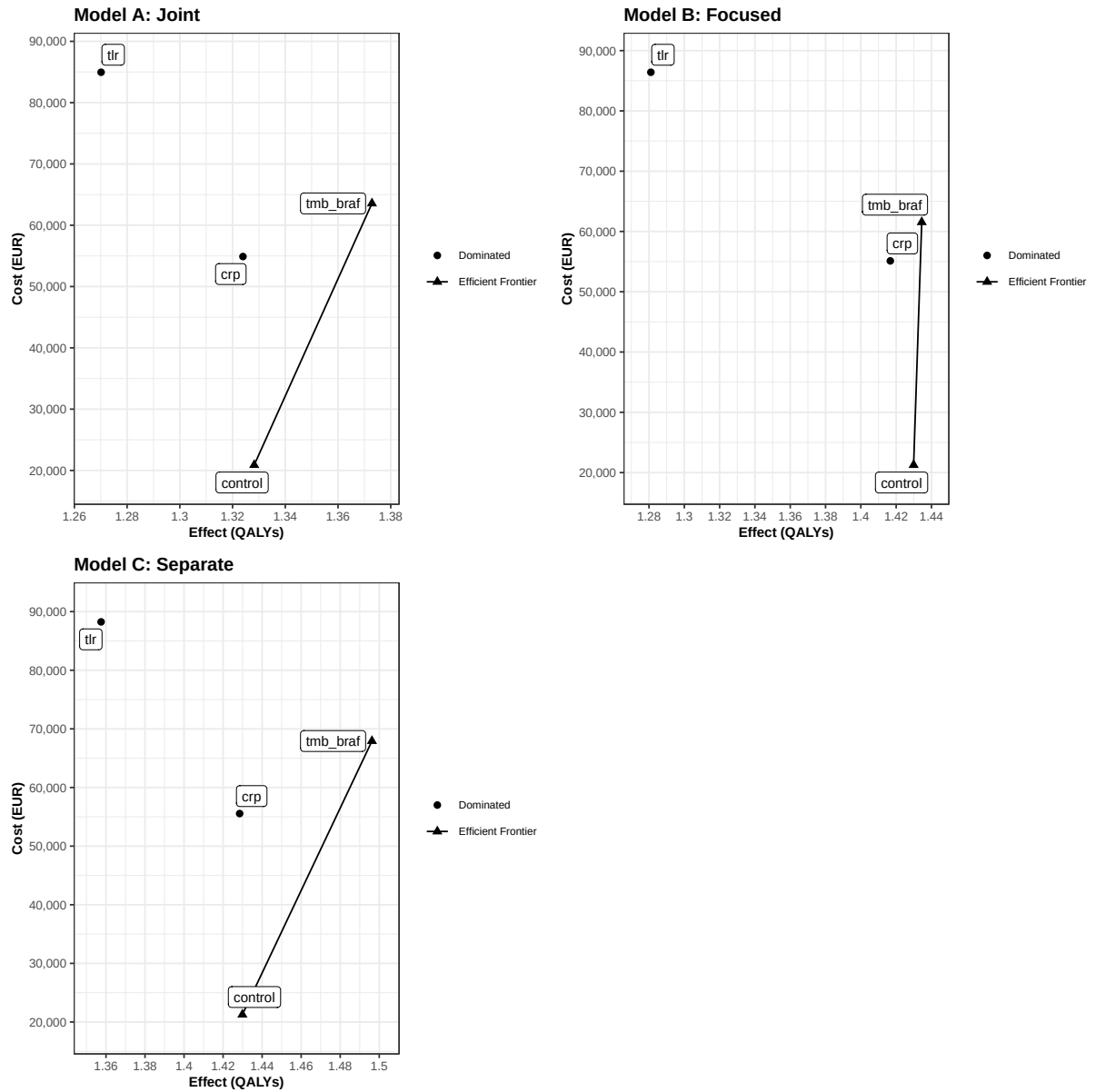


Figure 7: Cost-Effectiveness Planes by Model Structure - Scenario 2 (Alternative Utilities)

4.3 Probabilistic Sensitivity Analysis (Model A)

PSA results are presented for Model A (Joint) only.

Table: PSA Summary Statistics - Scenario 2 (Model A)

Table 14: PSA Summary - Scenario 2 (Model A: Joint, Alternative Utilities)

Strategy	meanCost	meanEffect	sdCost	sdEffect
control	21183.70	1.30	2864.38	0.26
crp	54980.02	1.35	9366.39	0.20
tlr	85454.92	1.31	17243.87	0.23
tmb_braf	63489.28	1.42	11247.09	0.26

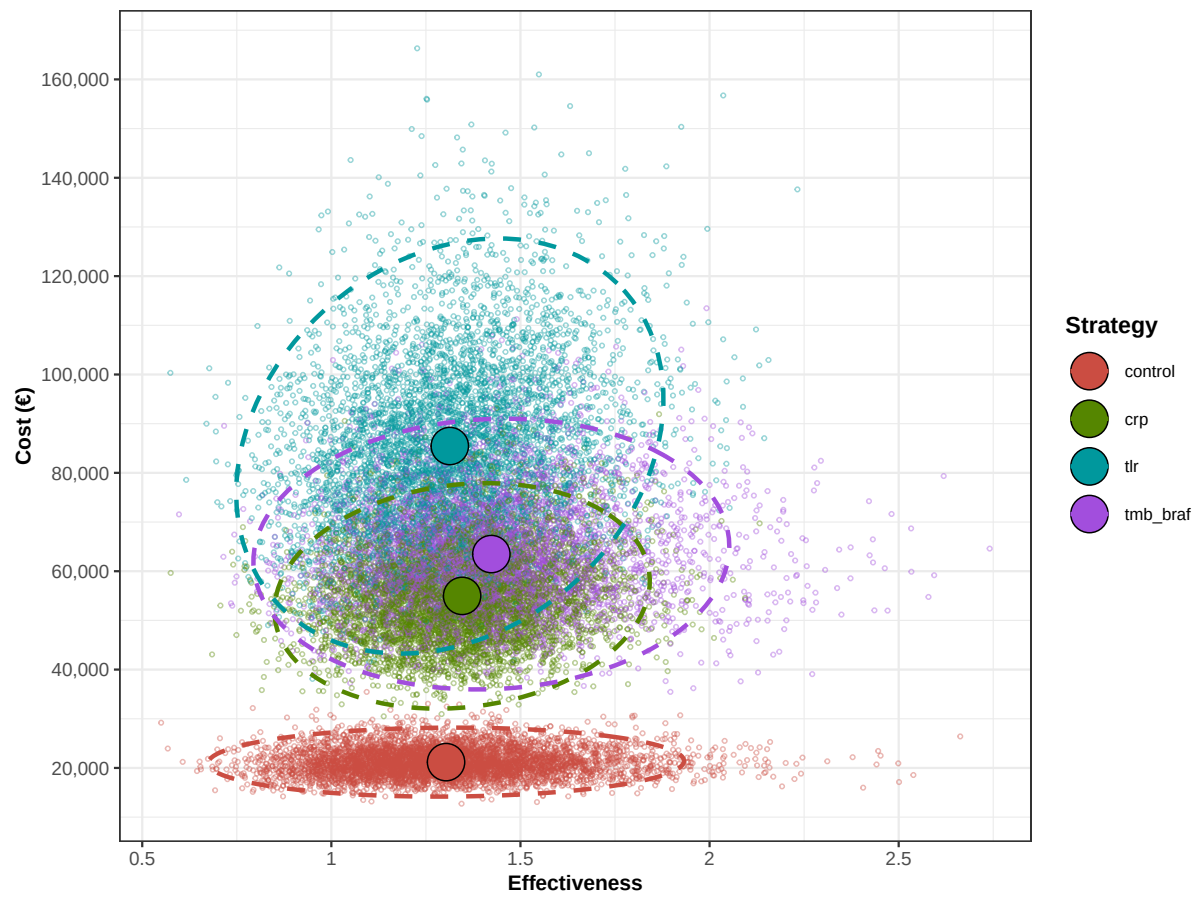


Figure 8: PSA Scatterplot - Scenario 2 (Model A: Joint, Alternative Utilities)

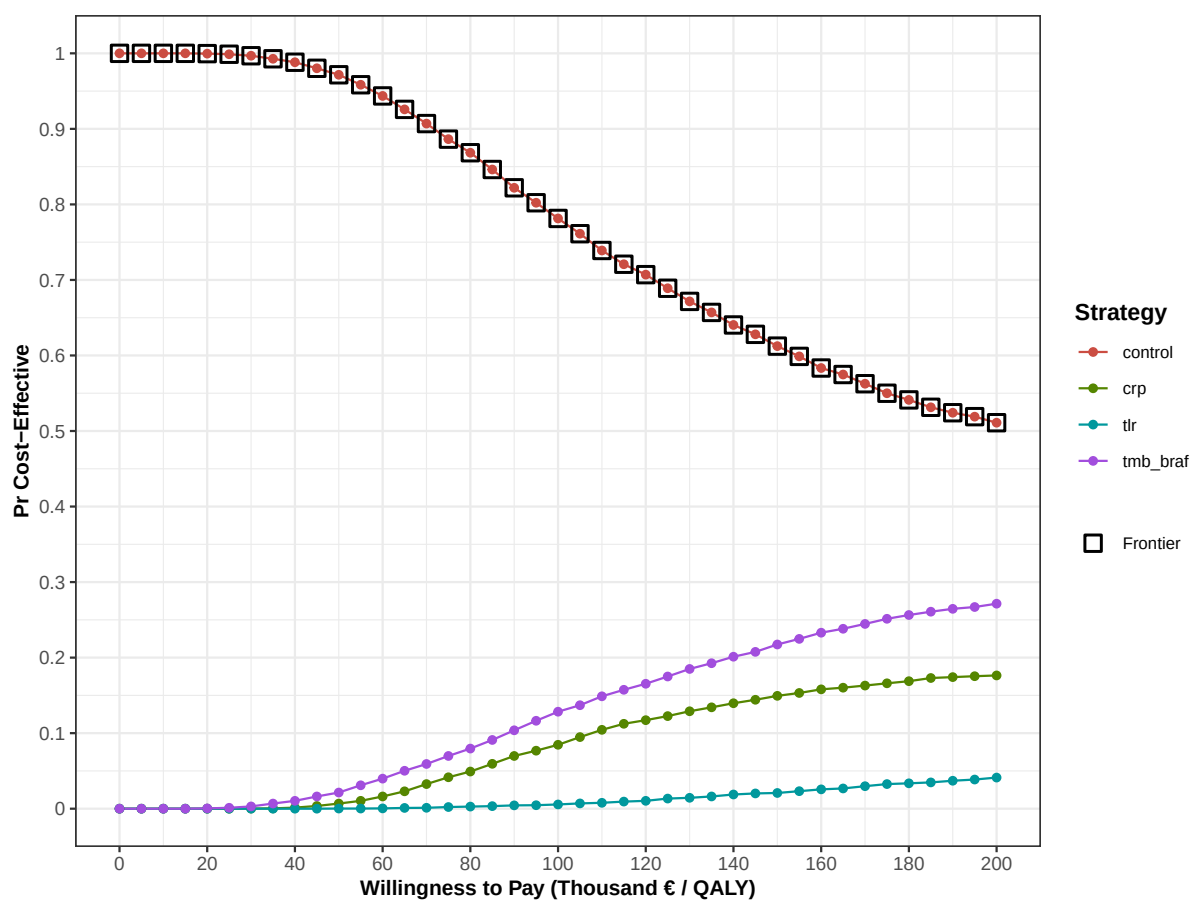


Figure 9: Cost-Effectiveness Acceptability Curve - Scenario 2 (Model A: Joint)

5 Comparative Analysis Across All Scenarios

5.1 Utility Parameter Summary

Table 15: Utility Parameter Values Across Scenarios

Scenario	u_np	u_p	Description
Base Case	0.9077	0.9005	Standard utility values from literature
Scenario 1 (Life Years)	1.0000	1.0000	No quality adjustment (all utilities = 1.0)
Scenario 2 (Alternative)	0.7300	0.5900	Lower/conservative utility estimates

5.2 Cross-Model, Cross-Scenario Optimal Strategy Comparison

Table 16: Optimal Strategy by Scenario and Model Structure (WTP = EUR 51,000)

Scenario	Model A: Joint	Model B: Focused	Model C: Separate
Base Case	Standard of Care	Standard of Care	Standard of Care
Scenario 1	Standard of Care	Standard of Care	Standard of Care
Scenario 2	Standard of Care	Standard of Care	Standard of Care

5.3 Model A Results Across Scenarios

The following tables and plots compare results across scenarios for Model A (Joint), which is the primary model structure.

Table 17: Cost-Effectiveness Results Comparison Across Scenarios (Model A)

Scenario	Strategy	Cost (EUR)	Effect	NMB (EUR)
Base Case	Standard of Care	\$20,891	1.846	\$73,254
	CRP-guided	\$54,894	1.783	\$36,045
	TLR-guided	\$84,959	1.702	\$1,859
	TMB/BRAF-guided	\$63,561	1.868	\$31,700
Scenario 1	Standard of Care	\$20,891	2.043	\$83,298
	CRP-guided	\$54,894	1.971	\$45,624
	TLR-guided	\$84,959	1.881	\$10,986
	TMB/BRAF-guided	\$63,561	2.065	\$41,777
Scenario 2	Standard of Care	\$20,891	1.328	\$46,848
	CRP-guided	\$54,894	1.324	\$12,627
	TLR-guided	\$84,959	1.270	-\$20,182
	TMB/BRAF-guided	\$63,561	1.373	\$6,456

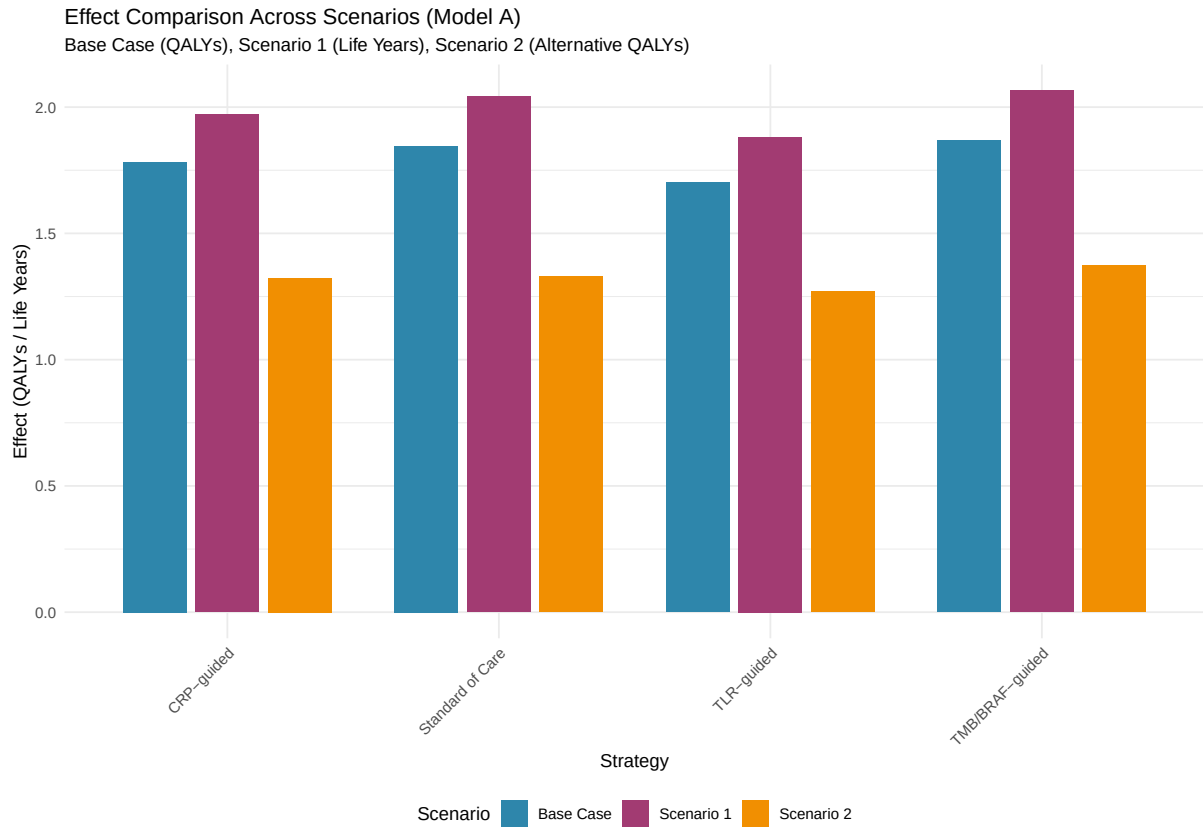


Figure 10: Comparison of Effects Across Scenarios by Strategy (Model A)

Table 18: ICER Comparison Across Scenarios - Model A (EUR per QALY/Life Year)

Strategy	Base Case	Scenario 1	Scenario 2
Standard of Care	Reference	Reference	Reference
TMB/BRAF-guided	\$1,949,087	\$1,894,792	\$955,489
CRP-guided	Reference	Reference	Reference
TLR-guided	Reference	Reference	Reference

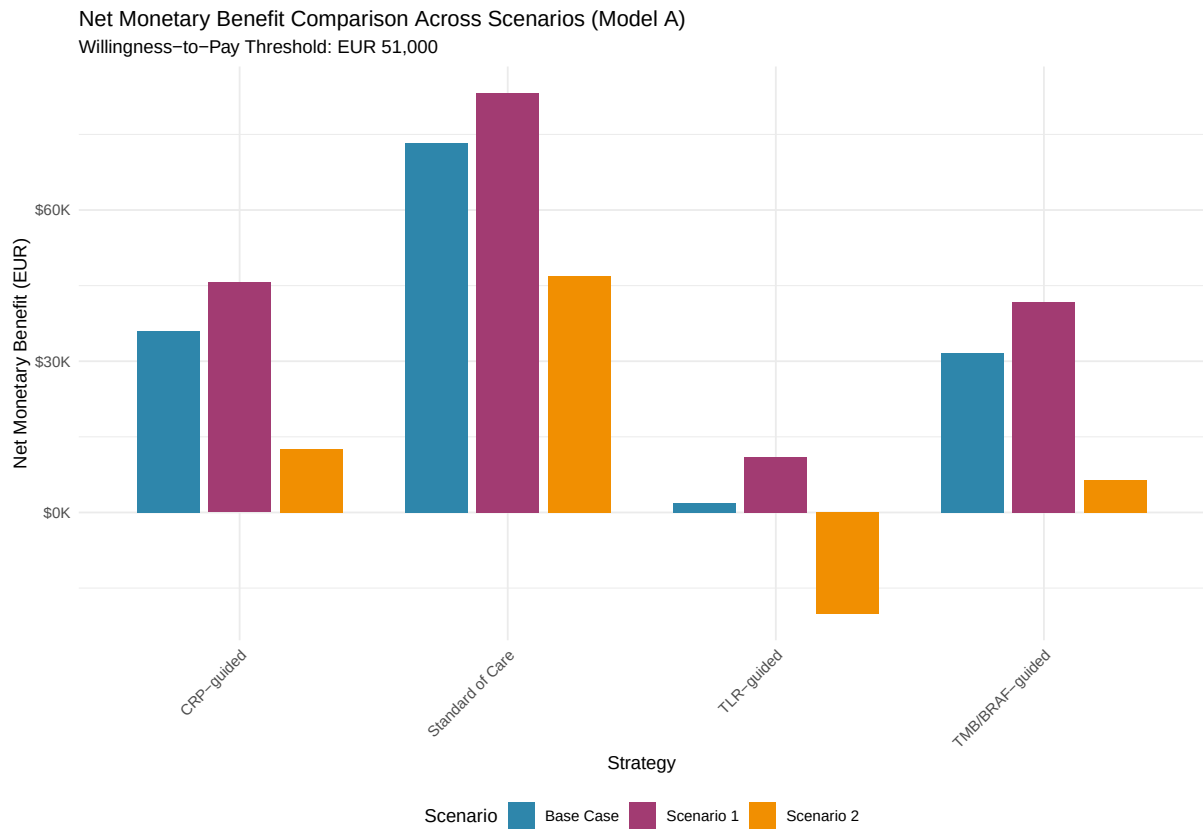


Figure 11: Net Monetary Benefit Comparison Across Scenarios (Model A, WTP = EUR 51,000)

6 Robustness Assessment

6.1 Cross-Model Consistency

Table 19: Robustness of Optimal Strategy Across Model Structures

Scenario	Model Consistency	Optimal Strategy(ies)
Base Case	Consistent	Standard of Care
Scenario 1	Consistent	Standard of Care
Scenario 2	Consistent	Standard of Care

Robust Conclusion: The optimal strategy is consistent across all three model structures within each scenario, suggesting the conclusions are robust to the choice of survival model specification.

6.2 Cross-Scenario Consistency

Table 20: Sensitivity of Optimal Strategy to Utility Assumptions by Model

Model Structure	Scenario Consistency	Optimal Strategy(ies)
Model A: Joint	Consistent	Standard of Care
Model B: Focused	Consistent	Standard of Care
Model C: Separate	Consistent	Standard of Care

7 Summary

This scenario analysis examined the impact of utility assumptions on the METIMMOX-1 cost-effectiveness model across three alternative model structures:

- **Model A (Joint):** All biomarkers and all treatment interactions in a single model
- **Model B (Focused):** Per-strategy formulas: CRP/TMB_BRAF with both interactions, TLR with TLR-only interaction
- **Model C (Separate):** Only one biomarker and its treatment interaction per model

Key findings:

1. **Deterministic results:** Multi-model comparison tables show how cost-effectiveness results vary across model structures within each scenario.
2. **Cross-model robustness:** The robustness assessment above indicates whether the optimal strategy is consistent across model structures.
3. **Scenario sensitivity:** The cross-scenario consistency analysis shows how sensitive conclusions are to utility assumptions for each model structure.
4. **PSA results:** Probabilistic sensitivity analysis was conducted for Model A (Joint) across all three scenarios to characterize decision uncertainty.

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